🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 10 · STANDARD 6.GR.2

Volume of Rectangular Prisms

Lesson 10-2 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the volume of a rectangular prism, including ones with fractional edge lengths, using base area × height.
Language Objective: I can explain my work using the words volume, rectangular prism, dimensions, and base area.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Volume | Rectangular prism | Cubic units | Dimensions | Net | Base area

👆 *Tap any word to see what it means and a picture.*

1. The amount of space inside a three-dimensional solid is its ________.

2. A solid with six rectangular faces is a ________.

3. The unit used to measure volume is ________.

4. The measurements of length, width, and height of a solid are its ________.

5. A flat pattern that folds into a solid is a ________.

6. The area of the bottom face of a prism, multiplied by height to find volume, is the ________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

COMPARE

Capsule A ($2 \times 1.5 \times 1$) and Capsule B ($3 \times 1 \times 1$) both have a volume of 3 ft^3. How did your multiplication produce the same volume from different dimensions?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Volume** — How much space is inside a solid shape.
  Volumen — Cuánto espacio hay dentro de una figura sólida.
- [ ] **Rectangular prism** — A solid box shape with six flat rectangle sides.
  Prisma rectangular — Una figura sólida en forma de caja con seis lados rectangulares.
- [ ] **Cubic units** — The units used to measure space inside, like cubic inches.
  Unidades cúbicas — Las unidades para medir el espacio interior, como pulgadas cúbicas.
- [ ] **Dimensions** — How long, how wide, and how tall a shape is.
  Dimensiones — Qué tan largo, ancho y alto es una figura.
- [ ] **Base area** — The area of the bottom of a solid. Volume = base area × height.
  Área de la base — El área del fondo de un sólido. Volumen = área de la base × altura.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

Both capsules have V = ____ ft³ because ____ × ____ × ____ = ____ × ____ × ____.
Ambas cápsulas tienen V = ____ ft³ porque ____ × ____ × ____ = ____ × ____ × ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: You still multiply length × width × height even when an edge is a fraction or decimal.

Evidence: Students explain the formula stays the same ($V = l \times w \times h = 2 \times 1.5 \times 1 = 3$ ft^3) and that a fractional edge just means multiplying with a fraction or decimal.

Reasoning: This evidence connects the definition of volume to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Both capsules have V = ____ ft^3 because ____ × ____ × ____ = ____ × ____ × ____.**
  Ambas cápsulas tienen V = ____ ft^3 porque ____ × ____ × ____ = ____ × ____ × ____.
- **The dimensions are different but the ____ is the same.**
  Las dimensiones son diferentes pero el ____ es el mismo.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Volume
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Cubic units
4. **Notes 4:** Dimensions
5. **Notes 5:** Net
6. **Notes 6:** Base area
7. **Watch for this mistake:** *A common mistake in Volume of Rectangular Prisms is rounding a fractional edge length to the nearest whole number before multiplying, instead of using the exact fraction or decimal. For a box that is 2 ft × 1.5 ft × 1 ft, a student might round 1.5 down to 1 and compute 2 × 1 × 1 = 2 ft^3, but the correct volume uses the exact edge length: 2 × 1.5 × 1 = 3 ft^3 — a full cubic foot larger. Before you submit, ask: 'Did I multiply using the exact fractional or decimal edge length, or did I round it off first?'*
8. **Try It 1:** A. Box A, because 50 cubic in is more than 48 cubic in — *Box A: 5 × 5 × 2 = 50 cubic in. Box B: 4 × 4 × 3 = 48 cubic in. 50 > 48, so Box A holds more.*
9. **Try It 2:** A. 18 cubic in — *V = 4 × 3 × 1.5 = 12 × 1.5 = 18 cubic in. A fractional edge does not change the steps.*